

Lecture 4: Elasticity

ECON 101
Fall 2020



Recall, from last lecture, the linear supply and demand:

$$Q_D = a - bP \quad \text{where } a, b, c, d > 0$$

$$Q_S = c + dP \quad \text{a, c encapsulate the determinants of D and S respectively}$$

$$Q_D(P_e) = Q_S(P_e), \text{ solve } P_e$$

$$P_e = \frac{a-c}{b+d}; \quad Q_e = \frac{ad+bc}{b+d}$$

For some applications, we would like to be explicit about some of the determinants of demand or supply.

For example:

let I = disposable income

P_s = price of a substitute

P_c = price of a complement.

$$Q_D = a - bP + \underbrace{fI + gP_s - hP_c}_{A = fI + gP_s - hP_c}$$

$$f, g, h > 0$$

$$P_e = \frac{a-c+A}{b+d}$$

$$Q_e = \frac{d(a+A) - bc}{b+d}$$

$$\text{Elasticity} : \frac{\% \text{ Change in one variable}}{\% \text{ Change in another variable}}$$

Shorthand notation: Δ denotes the change, so if we have variables x, y : $(x_1, y_1) \rightarrow (x_2, y_2)$

$$\Delta y = y_2 - y_1, \quad \Delta x = x_2 - x_1$$

$$\text{Elasticity} = \frac{\% \Delta y}{\% \Delta x} = \frac{\Delta y / y}{\Delta x / x} = \frac{\frac{\Delta y}{y_1}}{\frac{\Delta x}{x_1}} = \frac{\frac{y_2 - y_1}{y_1}}{\frac{x_2 - x_1}{x_1}}$$

Elasticity is a unitless quantity

Price Elasticity of Demand:

$$\epsilon_P^D = \frac{\% \Delta \text{Quantity Demanded}}{\% \Delta \text{Price}} = \frac{\Delta Q / Q}{\Delta P / P} = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

point
price-elasticity

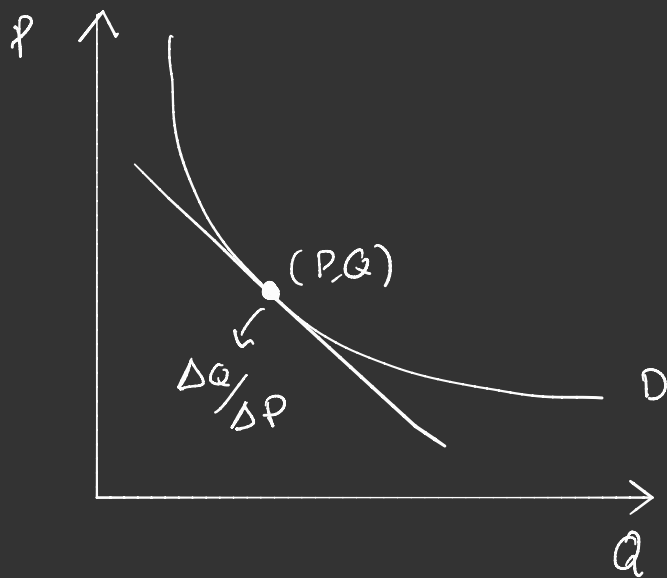
We have two components $\frac{\Delta Q}{\Delta P}$, $\frac{P}{Q}$

(1) $\frac{\Delta Q}{\Delta P}$ needs to be computed from the demand schedule function

(2) $\frac{P}{Q}$ we just need a point (P, Q) in the demand curve

How do we compute (1) $\frac{\Delta Q}{\Delta P}$

you need a second point (approximation), or you need information about $Q_D = Q_D(P)$ i.e. exact demand function



The computation of $\frac{\Delta Q}{\Delta P}$ is simplified a lot, for linear demand.

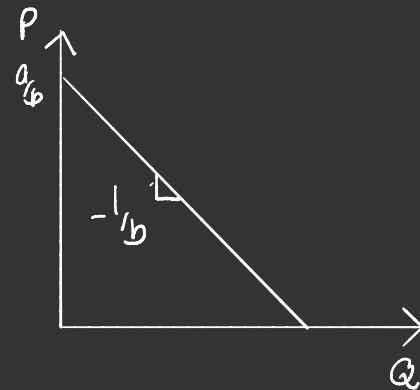
Suppose that $Q_0 = a - bP$; suppose that $P_1 \rightarrow P_2$

$$\left. \begin{array}{l} \text{At } P_1 \Rightarrow Q_1 = a - bP_1 \\ P_2 \Rightarrow Q_2 = a - bP_2 \end{array} \right\} \Rightarrow \Delta Q = Q_2 - Q_1 = a - bP_2 - (a - bP_1) = -b(P_2 - P_1)$$

$$\Delta P = P_2 - P_1$$

$$\Rightarrow \frac{\Delta Q}{\Delta P} = -b \frac{(P_2 - P_1)}{P_2 - P_1} = -b = \text{slope of the demand}$$

When graphing Q_0 : $P = \frac{a}{b} - \frac{1}{b} Q_0$
inverse demand



Thus for $Q_0 = a - bP$:

$$\epsilon_P^D = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q} = -b \cdot \frac{P}{Q} = \frac{-bP}{a-bP}$$

$\epsilon_P^D < 0$ always because of the Law of Demand,

Quantity is inversely related to price.

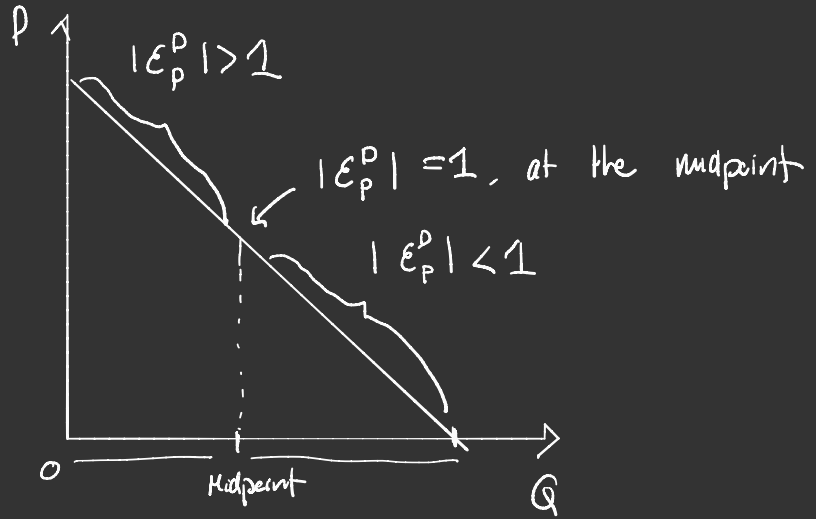
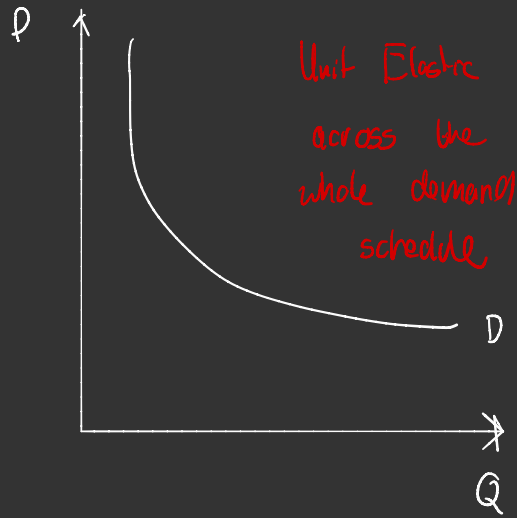
Thus, sometimes is useful to think about ϵ_P^D
in absolute terms, that is $|\epsilon_P^D|$

For any general demand curve:

$|E_P^D| > 1$: demand is **elastic**: $\frac{1\% \Delta Q}{1\% \Delta P} > 1 \Rightarrow |1\% \Delta Q| > |1\% \Delta P|$
quantity changes by more than the price change

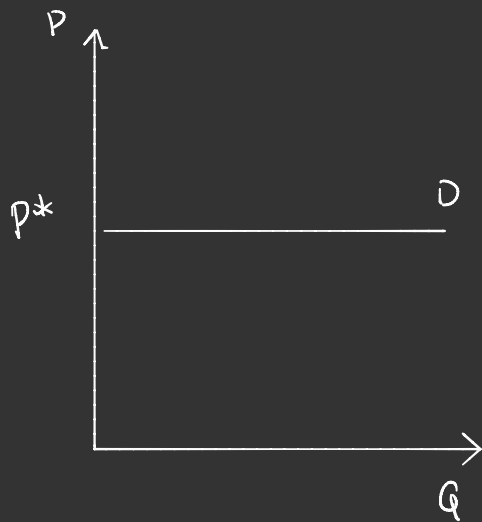
$|E_P^D| < 1$: demand is **inelastic**: $\frac{1\% \Delta Q}{1\% \Delta P} < 1 \Rightarrow |1\% \Delta Q| < |1\% \Delta P|$
quantity changes by less than the price change.

$|E_P^D| = 1$: demand is **unit elastic**: $\frac{1\% \Delta Q}{1\% \Delta P} = 1 \Rightarrow |1\% \Delta Q| = |1\% \Delta P|$
quantity changes equally to the change in price.



Linear Demand: has different areas of elasticity.

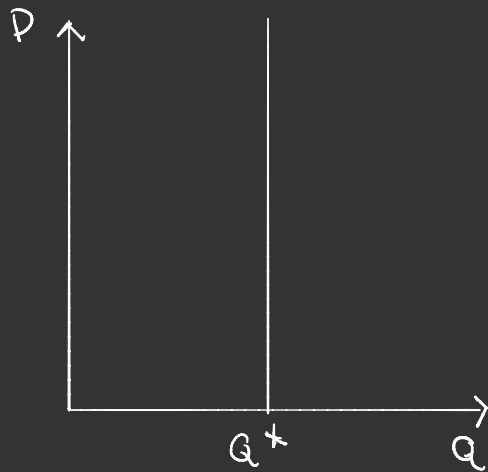
- First part is elastic: "responsive to price"
- Mid point is unit elastic
- Last-part is inelastic: "insensitive to price"



Perfectly Elastic Demand

$$p = p^* \rightarrow \frac{\Delta Q}{\Delta P} = \infty$$

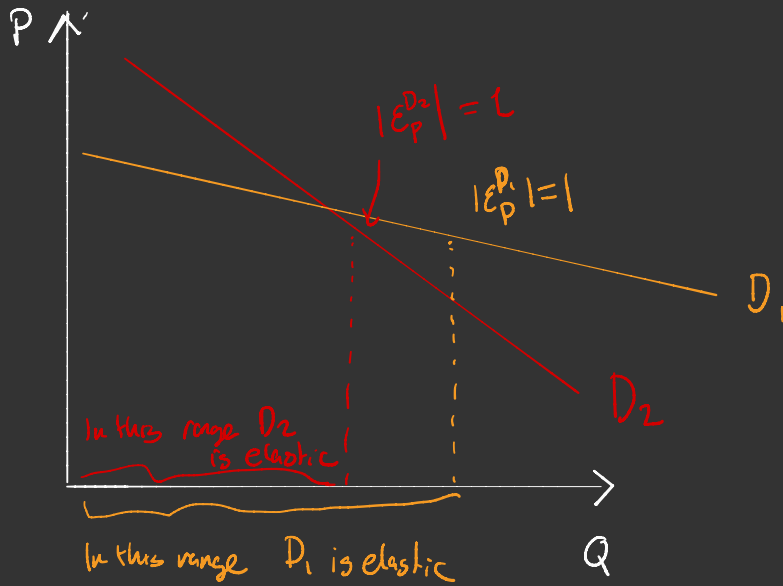
Extreme case, but goods that have many substitutes approach this case



$Q_D = Q^*$ constant with respect to price (it doesn't change)

$$\frac{\Delta Q}{\Delta P} = 0$$

Good Example, demand insuline



The flatter one is said to be more elastic with respect to the other

D_1 is more elastic than D_2

$$Q_D = a - bP + fI + gP_S - hP_C$$

$$\epsilon_I^D = \frac{\Delta Q}{\Delta I} \cdot \frac{I}{Q} = f \frac{I}{Q} = f \frac{I}{a - bP + fI + gP_S - hP_C}$$

$$\frac{\Delta Q}{\Delta I} = \frac{Q_2 - Q_1}{I_2 - I_1} = f$$

$$\epsilon_{Q_D, P_S} = \frac{\Delta Q_D}{\Delta P_S} \cdot \frac{P_S}{Q_D} = g \cdot \frac{P_S}{Q_D}$$

$$\frac{\Delta Q_D}{\Delta P_S} = g$$

$$\epsilon_{Q_D, P_C} = \frac{\Delta Q_D}{\Delta P_C} \cdot \frac{P_C}{Q_D} = -h \cdot \frac{P_C}{Q_D}$$

$$\frac{\Delta Q_D}{\Delta P_C} = -h$$

$$Q_S = c + dP$$

$$\epsilon_P^S = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q} = d \frac{P}{Q}$$

$$\frac{\Delta Q}{\Delta P} = d$$

Arc Elasticity of Demand:

Consider two points on the demand schedule, assume that you only know those two points: (P_1, Q_1) , (P_2, Q_2)

You don't know the exact relationship $Q_D = Q_D(P)$

$$\bar{\epsilon}_P^D = \frac{\frac{\Delta Q}{\bar{Q}}}{\frac{\Delta P}{\bar{P}}} = \frac{\frac{Q_2 - Q_1}{(Q_1 + Q_2)/2}}{\frac{P_2 - P_1}{\frac{P_1 + P_2}{2}}} = \frac{Q_2 - Q_1}{P_2 - P_1} \cdot \frac{(P_1 + P_2)/2}{(Q_1 + Q_2)/2} = \frac{Q_2 - Q_1}{P_2 - P_1} \cdot \frac{P_1 + P_2}{Q_1 + Q_2}$$

A similar concept can be defined for the arc elasticity of supply

$$\bar{\epsilon}_P^S = \frac{\Delta Q}{\Delta P} \cdot \frac{\bar{P}}{\bar{Q}}$$

How is the arc elasticity related to the point elasticity for linear demand curve?

For any linear demand $Q_D = a - bP$, $\frac{\Delta Q}{\Delta P} = -b$

$$\epsilon_P^D = -b \frac{P}{Q}, \quad \bar{\epsilon}_P^D = -b \frac{\bar{P}}{\bar{Q}}$$

When is the case $|\bar{\epsilon}_P^D| > |\epsilon_P^D| \Leftrightarrow$

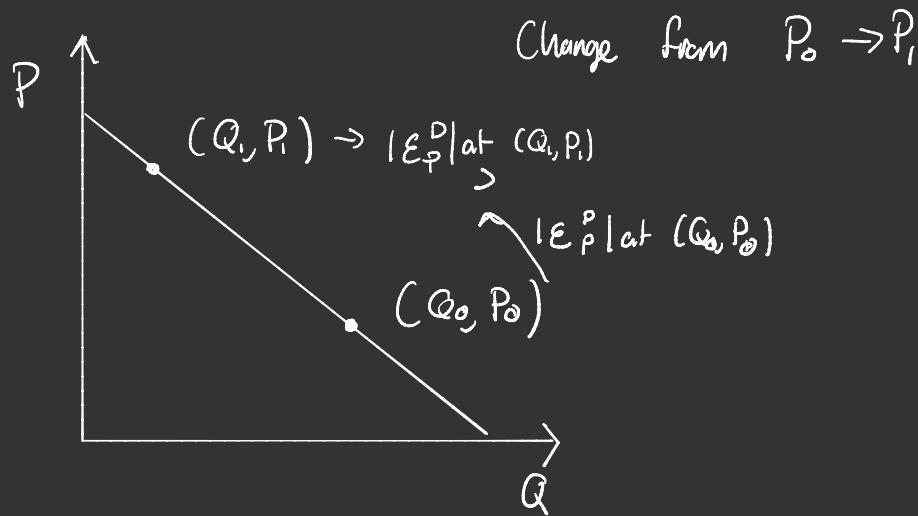
$$\Leftrightarrow b \frac{\bar{P}}{\bar{Q}} > b \frac{P}{Q} \Leftrightarrow \frac{\bar{P}}{\bar{Q}} > \frac{P}{Q} \quad (1), \text{ suppose that we were computing}$$

at points (P_0, Q_0) , (P_1, Q_1) , and we compute the point elasticity at (P_0, Q_0)

$$(1) \quad \frac{(P_0 + P_1)/2}{(Q_0 + Q_1)/2} > \frac{P_0}{Q_0} \Leftrightarrow \frac{P_0 + P_1}{Q_0 + Q_1} > \frac{P_0}{Q_0} \Leftrightarrow (P_0 + P_1)Q_0 > P_0(Q_0 + Q_1) \Leftrightarrow \frac{P_1 Q_0}{P_0 Q_0 + P_1 Q_0} > \frac{P_0 Q_0}{P_0 Q_0 + P_1 Q_0} \Leftrightarrow \frac{P_1}{Q_1} > \frac{P_0}{Q_0}$$

$$\frac{P_1}{Q_1} > \frac{P_0}{Q_0} \Leftrightarrow b \frac{P_1}{Q_1} > b \frac{P_0}{Q_0} \Leftrightarrow | \epsilon_P^D \text{ at point } (Q_1, P_1) | > | \epsilon_P^D \text{ at } (Q_0, P_0) |$$

$$| \overline{\epsilon}_P^D | > | \epsilon_P^D \text{ at point } (Q_0, P_0) | \Leftrightarrow | \epsilon_P^P | \text{ at point } (Q_1, P_1) > | \epsilon_P^D | \text{ at point } (Q_0, P_0)$$



Reasons to use arc elasticity :

- Not knowing the demand schedule
- Whenever the changes in price are large, it is better to use arc elasticity

Point Elasticity works best for small price changes

